CodeClash

CS455 Deliverable 3

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https://github.com/SrujanBhirud/CodeClash

5 November 2026

Problem and roles

- A contest platform: seats, submissions, a sandbox judge, and one hardening agent.
- Participants sit and submit. Setters author problems. The admin runs the contest.
- The agent proposes tests. A person publishes.
- Demo accounts use the password `codeclash`. Aarav submits. Meghna opens the saved run.

Requirements

- A seat exists only while `reserved < capacity`. 400 users, 200 seats, exactly 200 seated.
- A submission gets one verdict. A stale claim does not score it again.
- Six languages. The compiler can write. The run user cannot.
- Signup is a six-digit code. A participant who opens Admin is turned away.

Architecture

- React client, Express API, Socket.IO, Mongo replica set, Redis streams.
- Judge and agent are separate processes. The judge calls `docker run`.
- One machine. Mongo, Redis, and Mailpit stay on localhost.
- Caddy on port 80. No load balancer, no NAT gateway, no RDS.

Jira and GitHub

- Site `cs455-codeclash`. 58 issues. 39 Done. 1 In Progress. 18 To Do.
- CC-53 through CC-57 are Done. CC-53 is the compile permission, pull request 47.
- #40–#46 have no Jira key. They stay as merged. `main` is not rewritten.
- The row: CC-53, pull request 47, `languages.test.ts` green.

Agent boundary

- The worker may read the problem and run the sandbox.
- Publish, edit, and score are denied. A `publish` tool call is the audited deny in `ai.test.ts`.
- Gemini is optional: `gemini-3.6-flash`, thinking off. No key in git.
- The slot opens the saved run. The empty-provider file is the one that finishes in seconds.

Testing and deploy

- Coverage gate is 80% lines. Shared 95, agent 96, server 92, judge 94.
- Archive p95 was 49 ms after the aggregation (3 Oct). The 5 Nov rerun missed that threshold on a busy laptop and failed 0% of requests.
- A fork bomb stays inside the sandbox. Redis and Mongo health return, and the in-flight submission is judged once.
- No VM was started. This account's free tier lists only Glue, so a `t3.micro` would bill. The demo runs on this PC. `deploy/README.md` is the host recipe, and it ends with `stop-instances`.